

AWS → CLOUDFLARE MIGRATION · PROJECT DOCUMENTATION

Complete Project Documentation

AWS CloudFront, WAF & Shield Advanced → Cloudflare Migration

SaaS Edge Layer - AWS CloudFront/WAF/Shield → Cloudflare

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Audience: Client / CTO / Engineering Team

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1. Project Overview

This document describes the migration of a SaaS platform's public edge and security layer from AWS (Route 53, CloudFront, AWS WAF, Shield Advanced, and Firewall Manager) to Cloudflare. The application infrastructure stayed exactly where it was on AWS, only the edge layer moved. The result was the same level of protection appropriate to the client's risk, far simpler management, and a reduction in edge cost from over \$3,400/month to around \$200/month.

The client was a non-technical SaaS founder who was paying for enterprise-grade edge protection that had outgrown what their application actually needed at its current scale. The goal was to right-size that protection without touching the working application stack.

At a glance

Edge cost: ~\$3,400 / month → ~\$200 / month (about 94% lower)

Biggest driver: AWS Shield Advanced at \$3,000/month flat - built for high-value DDoS targets

New edge: Cloudflare Business - DNS, CDN, WAF, DDoS protection, SSL/TLS

Unchanged: ALB, ECS, RDS, Redis, VPC - the whole application stack stayed on AWS

Key safeguard: ALB locked to only accept traffic from Cloudflare IP ranges

1.1 What moved and what stayed

The design principle was simple, move only the public edge, and leave the application untouched. Cloudflare sits in front of everything and proxies traffic to the AWS ALB as the origin.

Moved to Cloudflare	Stayed on AWS
Public DNS (from Route 53)	Application Load Balancer (ALB)
CDN / caching (from CloudFront)	ECS workloads
WAF rules (from AWS WAF)	RDS and Redis
DDoS protection (from Shield Advanced)	VPC, security groups, internal networking
SSL/TLS and traffic visibility	-

2. Client Problem & Cost Analysis

The client came in confused about their AWS bill. They were paying over \$3,400/month just for the edge and security layer before any application or database costs and could not see what each service was doing or why it was there. In their own words “my application is not that critical to be spending this much on security every month.”

The setup was not wrong; it made sense when it was first built for a company planning for rapid growth and wanting enterprise-grade protection from day one. But as the business evolved, the cost of that protection had outgrown the application's actual needs at its current scale.

2.1 Where the edge cost came from

Service	Role	Cost driver
Shield Advanced	Managed DDoS protection	\$3,000/month flat - fixed fee regardless of traffic or attacks
CloudFront	CDN / caching	Usage-based edge delivery
AWS WAF	Web application firewall	Per-rule and per-request charges
Firewall Manager	Centralized policy management	Policy and config-tracking overhead
Route 53	Public DNS	Hosted zones and queries

The core issues

Shield Advanced is designed for high-value targets expecting sustained volumetric DDoS attacks. At \$3,000/month flat it dominated the bill, and for a growing startup it was overkill. Firewall Manager added further cost for centralized policy management that a single-account setup did not need.

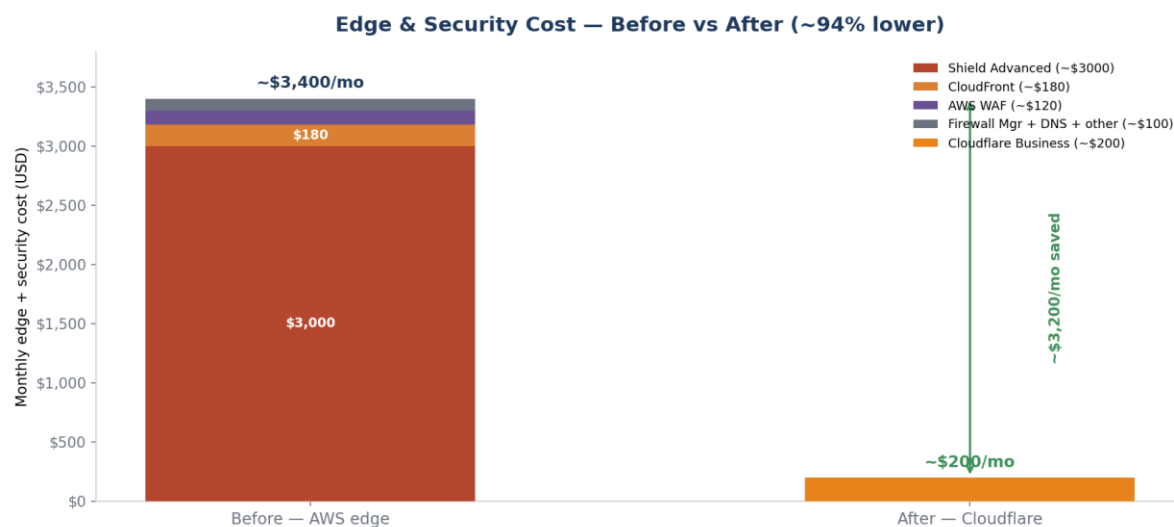


Figure 1 - Edge & security cost before vs after

3. Architecture Before - AWS Edge Stack

In the original architecture, all public traffic passed through a stack of AWS edge and security services before reaching the Application Load Balancer. Each service was managed in its own AWS console, which is part of why the setup felt opaque to a non-technical owner.

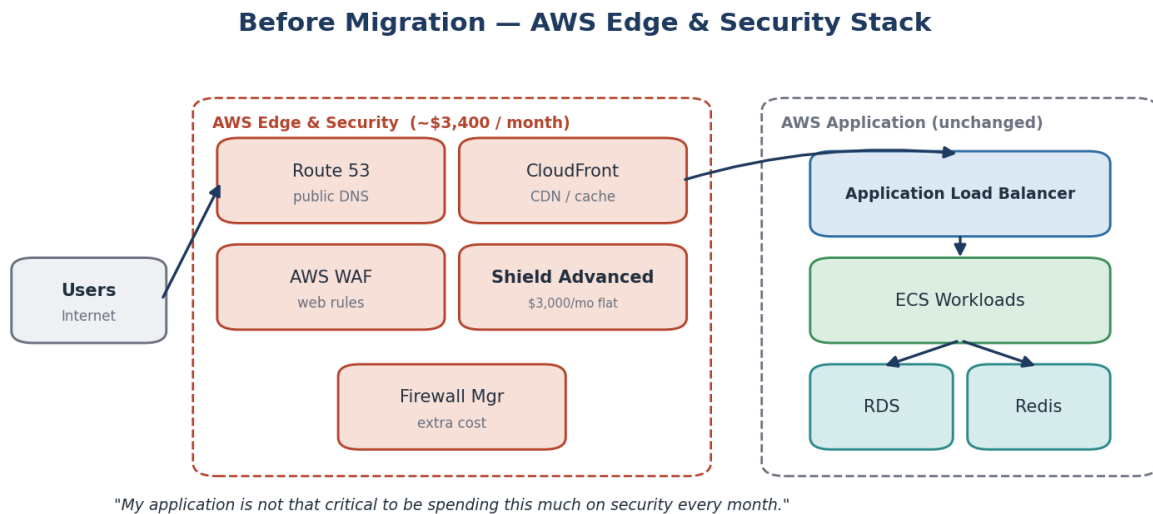


Figure 2 - AWS edge and security stack (before)

3.1 How traffic flowed

1. A user request resolved through Route 53 public DNS.
2. Traffic passed through CloudFront (CDN), AWS WAF (rules), and Shield Advanced (DDoS).
3. Firewall Manager centrally tracked the security policies.
4. Requests then reached the AWS ALB and the application behind it.

The application stack ALB, ECS, RDS, Redis were sound and stayed in place. The issue was purely the cost and complexity of the edge layer in front of it.

4. Migration Decision - Why Cloudflare

The recommendation was to move the public edge layer to Cloudflare and leave the application infrastructure exactly where it was. Cloudflare provides the same core edge capabilities DNS, CDN, WAF, and DDoS protection, bundled into a single plan, with one dashboard instead of six AWS consoles.

4.1 Why Cloudflare Business

The Cloudflare Business plan (around \$200/month) was chosen because it gave the client enough WAF control, custom rules, SSL/TLS features, and production-level options without the cost of an enterprise plan.

Need	How Cloudflare Business meets it
DDoS protection	Included as standard, no separate flat fee.
WAF & custom rules	Managed rulesets plus custom firewall rules.
CDN / caching	Global CDN with configurable cache rules.
SSL/TLS	Full Strict mode for end-to-end encryption.
Visibility	One dashboard for traffic, threats, and cache performance.

Right-sizing the protection

This was not about removing security, it was about matching the level of protection to the client's actual risk. Cloudflare Business covers a growing SaaS platform's needs at a fraction of the cost of Shield Advanced.

5. Migration Strategy

The migration was planned to be careful and reversible. The most delicate part was WAF rule parity, so that came first and was given the most attention. Nothing touched production until everything had been verified on a sandbox.

Migration Strategy — Phased & Reversible



Everything verified on a sandbox before any real traffic moved; rollback was available at each step.

Figure 3 - Phased, reversible migration approach

Phase	What happened
1 - Map	Map every existing AWS WAF rule to its Cloudflare equivalent (parity first).
2 - Build	Set up the Cloudflare zone, import DNS, configure SSL/TLS, cache, and WAF.
3 - Test	Verify DNS, caching, WAF, SSL, and origin routing on a sandbox before any real traffic.
4 - Cutover	Move nameservers to Cloudflare, then lock the ALB to Cloudflare IP ranges.
5 - Decommission	Carefully unsubscribe Shield Advanced, then remove Firewall Manager and CloudFront.

Why parity comes first

WAF rule parity is where migrations like this go wrong if rushed. Every IP list, rate limit, country rule, bot rule, and custom rule was mapped and verified before cutover.

6. Implementation & Configuration

Implementation followed the phased plan. Each area was configured in Cloudflare and verified on a sandbox before production traffic moved.

6.1 WAF rule mapping

Every existing AWS WAF rule was recreated in Cloudflare: managed rule groups, custom rules, rate limiting, IP allow/block lists, and country-based and bot rules. This was the most careful part of the work and is documented in full in Supporting Document 2 - WAF Rule Mapping.

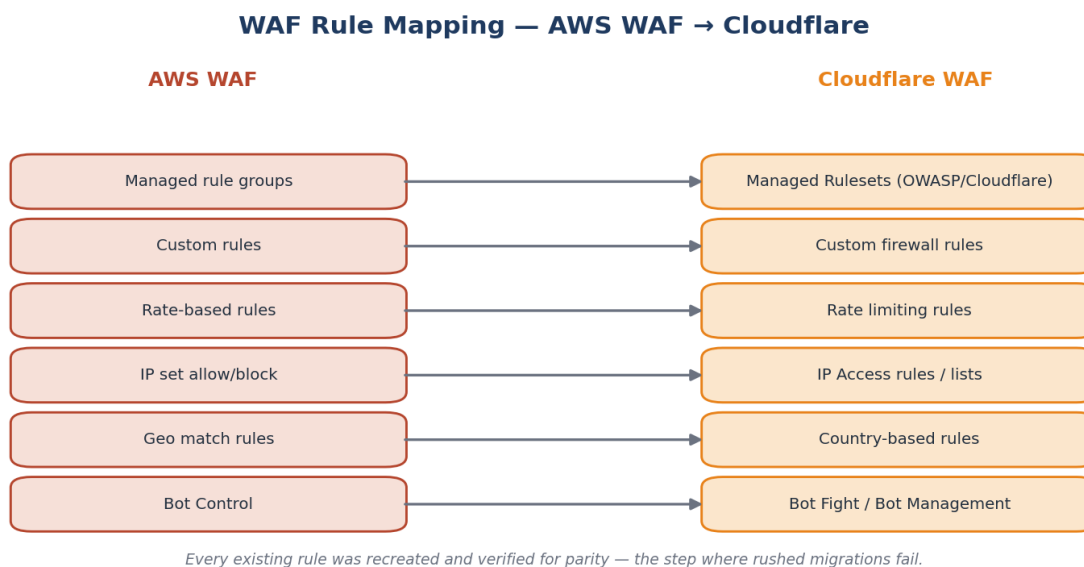


Figure 4 - AWS WAF rules mapped to Cloudflare equivalents

6.2 Zone, DNS & SSL/TLS

The Cloudflare zone was created, DNS records were imported, and SSL/TLS was configured in Full Strict mode to keep traffic encrypted end-to-end between the user, the Cloudflare edge, and the AWS ALB origin.

- Cloudflare zone set up and DNS records imported to match the existing configuration.
- SSL/TLS set to Full Strict so certificates are validated all the way to the origin.
- CDN cache rules configured to match the caching behavior CloudFront had provided.

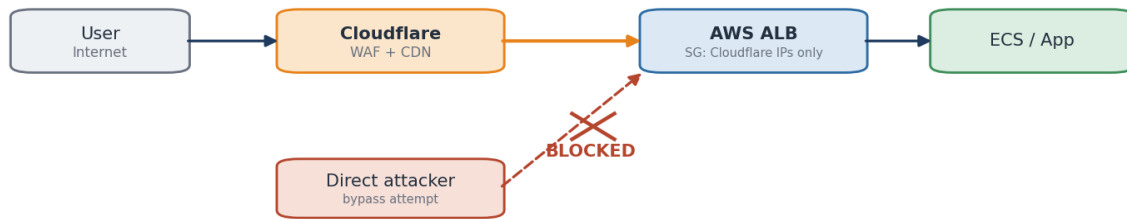
6.3 Sandbox testing

Before touching production, everything was tested on a sandbox: DNS records, CDN caching rules, WAF rulesets, SSL, and origin routing were all verified while real traffic still flowed through AWS.

6.4 Origin protection

After cutover, the ALB security group was locked down to accept traffic only from Cloudflare IP ranges. This is the step many people miss — without it, an attacker could bypass Cloudflare and hit the AWS origin directly.

Origin Protection — Locking the ALB to Cloudflare



The ALB security group only accepts Cloudflare IP ranges, so no one can bypass Cloudflare and hit the origin directly.

Figure 5 — Locking the ALB origin to Cloudflare IP ranges

7. Cutover & Post-Migration Monitoring

Once sandbox testing was clean, the nameservers were moved to Cloudflare. The decommissioning of AWS services was then handled carefully, in order, with the most cost-sensitive service treated with the most care.

7.1 Cutover sequence

5. Move the nameservers to Cloudflare so it becomes the authoritative edge.
6. Lock the ALB security group to Cloudflare IP ranges only.
7. Unsubscribe Shield Advanced, working around the client's billing and renewal window.
8. Decommission Firewall Manager and CloudFront one by one.

Handling the Shield Advanced commitment

Shield Advanced carries a subscription commitment, so it was not removed blindly. The cancellation was timed around the client's billing and renewal window rather than incurring an early-termination penalty.

7.2 Monitoring after cutover

For a week after cutover, traffic was watched closely to confirm real users were unaffected.

- Cloudflare analytics for traffic patterns and edge behaviour.
- WAF events to confirm rules were firing correctly.
- ALB access logs to confirm traffic was arriving only via Cloudflare.
- Application error rates to catch any regression early.

8. Final Outcome

The edge layer was right-sized to the client's actual needs, with no change to the working application stack on AWS.

Area	Before	After
Edge + security cost	~\$3,400 / month	~\$200 / month
DDoS protection	Shield Advanced (\$3,000 flat)	Included with Cloudflare
WAF	AWS WAF	Cloudflare WAF (full parity)
DNS	Route 53	Cloudflare DNS
CDN	CloudFront	Cloudflare CDN
Management	Six AWS consoles	One Cloudflare dashboard
Origin security	Open to direct access	ALB locked to Cloudflare IPs
Application stack	AWS (ALB, ECS, RDS, Redis)	Unchanged, still on AWS

Result

The client got the right level of protection for their current risk, simpler rule management, one dashboard for traffic, threats, and cache performance and an edge bill reduced from over \$3,400/month to around \$200/month, with the application stack left fully intact.

8.1 Deliverables handed to the client

- Full cost audit report breaking down each AWS edge service and its contribution to the bill.
- WAF rule mapping document, every AWS WAF rule mapped to its Cloudflare equivalent with notes.
- Cloudflare account setup: DNS, CDN cache rules, WAF rulesets, rate limiting, firewall rules, SSL/TLS Full Strict.
- Sandbox test report confirming caching, WAF, SSL, and origin routing before cutover.
- Updated ALB security group restricted to Cloudflare IP ranges only.
- Step-by-step cutover runbook with rollback procedure.
- Post-migration monitoring report (Cloudflare analytics, WAF events, ALB logs, error rates).
- Handover document explaining the full setup and how to manage it going forward.